

RACHANA BONIGALA

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Technical Skills

Programming Languages: Python, C++, C, SQL

Web & Backend Development: FastAPI, HTML, CSS, Object-Oriented Programming

Core CS Concepts: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks

AI & Automation: OpenAI API, LangChain, Prompt Engineering, Zapier

Developer Tools: Git, GitHub, VS Code, Notion

Education

B.Tech – Artificial Intelligence

2024 – 2028

Sardar Vallabhbhai National Institute of Technology (SVNIT)

Surat, Gujarat

– CGPA: 7.10 / 10 (up to 4th semester)

– Relevant Coursework: Data Structures & Algorithms, Artificial Intelligence, Operating Systems, Computer Networks, Database Management Systems

Class XII – MPC (Board of Intermediate Education, AP)

2022 – 2024

Andhra Pradesh

India

– Scored 963/1000 (96.3%)

Class X – Indian Certificate of Secondary Education (ICSE)

2021 – 2022

Andhra Pradesh

India

– Scored 90.5%

Projects

Life Flow – Home Management System | *Python, Django, SQLite, HTML, CSS*-GitHub

2025

- Built a full-stack Django web application for expense tracking, scheduling, and inventory management, applying object-oriented design across multiple interconnected modules.
- Designed and normalized a SQLite relational database schema, writing SQL queries to efficiently manage cross-module relationships and data integrity.
- Implemented user authentication, role-based access control, and session management, ensuring secure, scalable access to application features.

AI Research Assistant Agent | *Python, FastAPI, LangChain, REST APIs, JavaScript*-GitHub

2026

- Engineered a full-stack application using FastAPI, LangChain, and Groq Llama-3.3-70B to automate topic research and knowledge synthesis end-to-end.
- Built and integrated RESTful API endpoints with asynchronous JavaScript requests and JSON-based data exchange between frontend and backend.
- Designed a multi-stage retrieval pipeline that generates optimized queries, retrieves live source data, and produces grounded, citation-backed summaries.

Say Cheese – Browser Photobooth App | *JavaScript, HTML, CSS, Web APIs*-GitHub

2025

- Developed a browser-based photobooth application using JavaScript and Web APIs, including MediaDevices and Canvas, for real-time media processing.
- Implemented real-time image capture, image filters, and downloadable photo generation entirely client-side.
- Applied object-oriented programming principles to structure filter logic, application state, and user interaction handling.

Research Experience

Explainable AI (XAI) for depression detection | *Python, SHAP, LIME*

2026 – Present

- Investigating SHAP and LIME explainability frameworks on NLP models to generate interpretable, clinician-facing diagnostics for mental health assessments.
- Analyzing token- and phrase-level feature importances in clinical text datasets to connect black-box model outputs to real-world clinical workflows.

Achievements & Extracurriculars

- **Competitive Programming:** Solved 90+ problems on LeetCode, applying data structures, algorithms, and analytical problem-solving under time constraints
- **School People Leader (SPL):** Led student initiatives and represented peers in faculty discussions, building leadership, teamwork, and communication skills
- **Podcast Hosting & Content Creation:** Independently created, scripted, and hosted a podcast exploring psychology and student life, managing research, production, and audience engagement end-to-end. [Podcast Link]
- **Language Learning:** Actively learning German on Duolingo with a 290+ day streak, demonstrating consistency, discipline, and self-driven learning